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ACADEMIC VAULT

UNIZIK FRESHER EDITION | 2026

FEG 101: Engineering Mathematics I - Functions, Calculus and Vectors

The mathematical toolkit that every first-year engineering course assumes you already have

FACULTY PRIMER

Faculty Primer

Faculty of Engineering
Engineering (Common 100L Courses) | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

START HERE

Why this pack exists

Engineering students fail FEG 101 for two reasons: weak algebra and rushing calculus without understanding limits. This pack rebuilds the algebra foundations, explains differentiation and integration through physical meaning, and drills vector operations with engineering-flavoured worked examples.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Manipulate algebraic expressions, indices, logarithms and partial fractions without calculator dependence.
- Evaluate limits and differentiate standard functions, including product, quotient and chain rules.
- Integrate by substitution and by parts, and apply definite integrals to area and work problems.
- Add, resolve and take dot and cross products of vectors in two and three dimensions.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

Repair the algebra first

Most 'calculus problems' are actually algebra problems wearing a disguise. Fix these and the rest of the course becomes manageable.

- Indices: a^m x a^n = a^(m+n); a^(m/n) is the n-th root of a^m.
- Logarithms: log(ab) = log a + log b; log(a^n) = n log a. Know the change-of-base rule.
- Factorise before you substitute. A common factor often cancels an apparent difficulty.
- Partial fractions: split a rational expression into simpler denominators before integrating.
- Quadratic formula and completing the square reappear in optimisation and in dynamics.

> **No-calculator drill**
> Spend 15 minutes weekly evaluating simple expressions by hand. Engineering exams reward speed and accuracy, and both come from fluency, not from the calculator.

Limits and differentiation

The derivative is an instantaneous rate of change. Understand that sentence and the rules become tools rather than magic.

The derivative of f at x is the limit of the change in f divided by the change in x as the change approaches zero. That is why velocity is the derivative of displacement and current is the derivative of charge.

Learn the standard derivatives, then the combination rules: product, quotient and chain. The chain rule is the one students under-practise, and it appears in almost every engineering application.

- d/dx of x^n is n x^(n-1); of sin x is cos x; of cos x is -sin x; of e^x is e^x; of ln x is 1/x.
- Product rule: (uv)' = u'v + uv'.
- Quotient rule: (u/v)' = (u'v - uv') / v^2.
- Chain rule: differentiate the outside, then multiply by the derivative of the inside.
- Use the first derivative for increasing/decreasing behaviour and the second for concavity.

Integration and vectors

Integration reverses differentiation and accumulates quantity. Vectors describe quantities with both size and direction.

- A definite integral gives signed area. Always sketch the curve to check whether your answer is plausible.
- Substitution works when you can see the derivative of an inner function in the integrand.
- Integration by parts follows the product rule; choose u so that differentiating it simplifies the expression.
- Resolve a vector into components: Fx = F cos(theta), Fy = F sin(theta).
- Dot product gives work and the angle between vectors; cross product gives torque and a perpendicular vector.

> **Units are a check**
> If you are integrating a velocity in m/s over time in seconds, the answer must be in metres. If your units are wrong, your method is wrong.

Functions, limits and continuity

Engineering mathematics models physical systems. A function describes how one quantity depends on another; its domain and range describe what is physically meaningful. Choosing the right function - linear, exponential, sinusoidal - is a modelling decision before it is a mathematical one.

Limits describe behaviour near a point rather than at it, which is what makes calculus possible. Continuity requires the limit to exist, the function to be defined there, and the two to agree. Engineering functions are continuous except at switches, impacts and boundaries - exactly where careful analysis matters.

Differentiation gives the rate of change: velocity from displacement, current from charge. Integration accumulates: distance from velocity, work from force. The fundamental theorem connects them.

Differentiation and its applications

The power, product, quotient and chain rules become mechanical with practice. The chain rule is most often misused because the inner function must be identified explicitly. Implicit differentiation handles relations that cannot be written as y equals f of x.

Optimisation follows a pattern: find stationary points by setting the derivative to zero, classify them with the second derivative, and check the endpoints. Word problems require translating a situation into a function of one variable, usually by eliminating a constraint.

Related rates and small-error estimation interpret the derivative as a sensitivity: how much does the output change per unit change in the input?

> **Draw the graph first**
> A sketch reveals how many stationary points exist and roughly where. Then calculate, and check that the algebra agrees.

Integration and linear algebra

Not every function has an elementary antiderivative. Standard techniques - substitution, by parts, partial fractions, trigonometric identities - are pattern recognition. Definite integrals give areas, volumes, centroids, work and average values.

Taylor series approximate functions by polynomials, which is how calculators and simulations work. The error term tells you how good the approximation is, and engineering depends on knowing that.

Linear algebra supplies the language of systems: matrices represent transformations, determinants test invertibility, eigenvalues describe modes and stability. Elimination is the basis of structural, circuit and control analysis.

WORKED EXAMPLES

Do it once, slowly

These are not model answers to copy. They show the reasoning a strong student would use, step by step, so you can repeat the process on a question you have not seen.

1. Maximising an area

A rectangular field is fenced with 100 m against an existing wall. Find the dimensions giving maximum area.

1. Let width be x and length y, with the wall forming one side.
2. Constraint: 2x + y = 100, so y = 100 - 2x.
3. Area: A = x(100 - 2x) = 100x - 2x squared.
4. Differentiate: 100 - 4x = 0, so x = 25.
5. Then y = 50 and area is 1,250 square metres.
6. Second derivative is negative, confirming a maximum; endpoints give zero.

> **What to carry forward**
> Constraint, single-variable function, derivative, classification.

2. A definite integral for work

A force F = 3x newtons acts along a line. Find the work done from x = 0 to 4 m.

1. Work is the integral of force over displacement.
2. Set up the integral of 3x dx from 0 to 4.
3. Antiderivative: 1.5 x squared.
4. Evaluate: 1.5 x 16 = 24 joules.
5. Check units: newtons times metres gives joules.
6. Sanity-check against the average force of 6 N over 4 m.

> **What to carry forward**
> Set up from the physical definition, then check units and magnitude.

EXAM TECHNIQUE

Questions with model answers

Written in the shape your course is usually examined, with the marking logic spelled out. Attempt each one under time before you read the outline.

Q1. Find and classify the stationary points of f(x) = x cubed - 3x.

15 marks

1. Differentiate: 3x squared - 3.
2. Set to zero: x = 1 or x = -1.
3. Second derivative: 6x, positive at 1 and negative at -1.
4. Minimum at x = 1, maximum at x = -1.
5. State the function values.

> **Where marks are lost**
> Failing to classify with the second derivative or to give the values.

Q2. Evaluate the integral of x e to the x dx.

10 marks

1. Integration by parts with u = x and dv = e to the x dx.
2. Apply the formula: x e to the x minus the integral of e to the x dx.
3. Result: e to the x (x - 1) plus a constant.
4. Differentiate to check.

REFERENCE

Glossary

Use these definitions in your own sentences; a copied definition rarely earns full marks.

Limit. The value a function approaches as the input approaches a point.

Continuity. The property that a function's limit equals its value at a point.

Derivative. The instantaneous rate of change of a function.

Stationary point. A point where the derivative is zero.

Chain rule. The rule for differentiating a function of a function.

Integral. Accumulation of a quantity, computed as an antiderivative or area.

Definite integral. An integral evaluated between limits, giving a numeric quantity.

Taylor series. A polynomial approximation of a function around a point.

Matrix. A rectangular array representing a linear transformation.

Determinant. A scalar indicating invertibility and scaling of a matrix.

Eigenvalue. A scalar by which an eigenvector is scaled.

Optimisation. Finding a maximum or minimum subject to constraints.

Your four-week plan

Week 1 - Functions

Limits.

- Evaluate ten limits.
- Test five functions for continuity.
- Sketch three graphs.

Week 2 - Differentiation

Rules.

- Differentiate twenty functions.
- Apply the chain rule to ten composites.
- Solve two optimisation problems.

Week 3 - Integration

Techniques.

- Integrate by substitution and parts.
- Evaluate four definite integrals.
- Compute one area and one volume.

Week 4 - Linear algebra

Systems.

- Solve a 3x3 system.
- Compute three determinants.
- Find two eigenvalues.

Where to read more

Entry-level texts you can find in the university library or a departmental reading list. Confirm editions with your lecturer.

- Any engineering mathematics textbook covering calculus and linear algebra.
- A problem set with worked solutions.
- A mathematical methods text covering series approximations.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. Differentiate y = (3x^2 + 1)^5.

Answer: Use the chain rule: dy/dx = 5(3x^2 + 1)^4 x 6x = 30x(3x^2 + 1)^4.

2. Integrate the integral of 2x cos(x^2) dx.

Answer: Let u = x^2, du = 2x dx. The integral becomes the integral of cos u du = sin(x^2) + C.

3. A force of 10 N acts at 30 degrees to the horizontal. Resolve it.

Answer: Fx = 10 cos 30 = 8.66 N; Fy = 10 sin 30 = 5 N.

4. Evaluate the limit of (x^2 - 4)/(x - 2) as x approaches 2.

Answer: Factorise: (x - 2)(x + 2)/(x - 2) = x + 2, so the limit is 4.

5. Why is the second derivative useful in optimisation?

Answer: It distinguishes a maximum from a minimum: negative second derivative indicates a maximum, positive indicates a minimum.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Rewrite the indices, logarithm and partial fraction rules on one page without notes.
- [ ] Do 15 differentiation problems covering product, quotient and chain rules.
- [ ] Do 10 integration problems split evenly between substitution and parts.
- [ ] Resolve five force vectors into components and recombine them.
- [ ] Sketch two curves and estimate the area before integrating to check your answer.
- [ ] Attempt one past FEG 101 question under timed conditions and mark it honestly.
- [ ] List every topic you could not solve and take the list to your lecturer or tutorial group.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date:

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